YEAR 11 PHYSICS, AQA GCSE, LESSON 2 OF 10

Atoms & Nuclear Radiation.

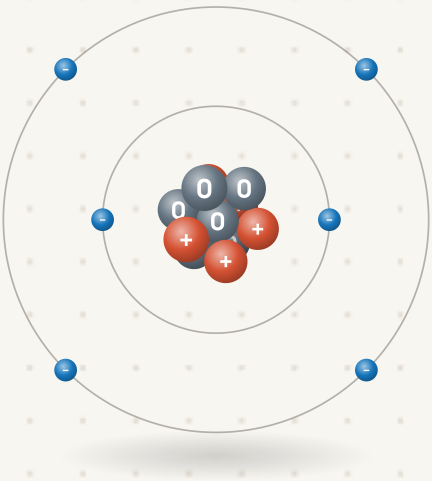
Lesson 02 Isotopes and ions

Carbon-12 and carbon-14 are both carbon, but only carbon-14 is radioactive. What is different about them?

IN THIS LESSON YOU WILL

- explain what isotopes are
- compare isotopes using nuclear notation
- describe how an atom becomes a positive ion
- say what changes, and what does not, when an ion forms

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lesson 2** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this lesson is highlighted)

01 Inside the atom

02 Isotopes and ions
Same element, different neutrons; atoms that lose outer electrons (slides 7–8)

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

02 Isotopes and ions.

AQA Physics 4.4.1.2; Combined Science 6.4.1.2

BIG QUESTION

Carbon-12 and carbon-14 are both carbon, but only carbon-14 is radioactive. What is different about them?

BY THE END YOU CAN

- explain what isotopes are
- compare isotopes using nuclear notation
- describe how an atom becomes a positive ion
- say what changes, and what does not, when an ion forms

isotope | ion | positive ion | outer electron | stable | unstable

SLIDES AND SIMULATIONS

Teaching deck, slides 7–8

Isotopes of carbon; a lithium atom becoming an ion.

panphy.app/gcsephy/decks/atoms-and-radiation/#7

RETRIEVE

Do now

From earlier lessons. No notes!

1 What is the relative charge of a neutron?

0

2 How many neutrons are in $^{31}_{15}P$?

31 − 15 = 16

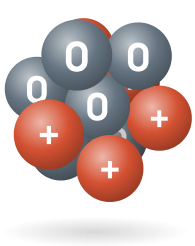
3 Why does an atom have no overall charge?

It has the same number of electrons as protons.

EXPLAIN

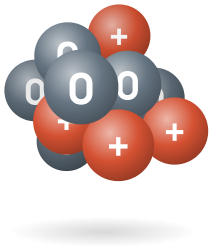
Same element, different atoms

Isotopes



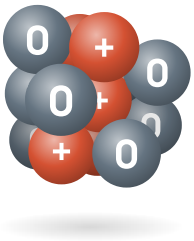
carbon-12

6 p, 6 n



carbon-13

6 p, 7 n



carbon-14

6 p, 8 n

Isotopes are atoms of the **same element** with **different numbers of neutrons**. They have the same number of protons, so the same atomic number, but different mass numbers. We name an isotope by its mass number: carbon-12, carbon-13, carbon-14.

Isotope	Symbol	Protons	Neutrons	Stable?
carbon-12	$^{12}_{6}C$	6	6	stable
carbon-13	$^{13}_{6}C$	6	7	stable
carbon-14	$^{14}_{6}C$	6	8	unstable: radioactive

! Isotopes: **same protons, different neutrons**. Some isotopes have an **unstable** nucleus: it gives out radiation. You will meet this in Lesson 5.

Hydrogen's three isotopes



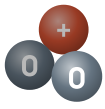
1 p, 0 n

hydrogen-1



1 p, 1 n

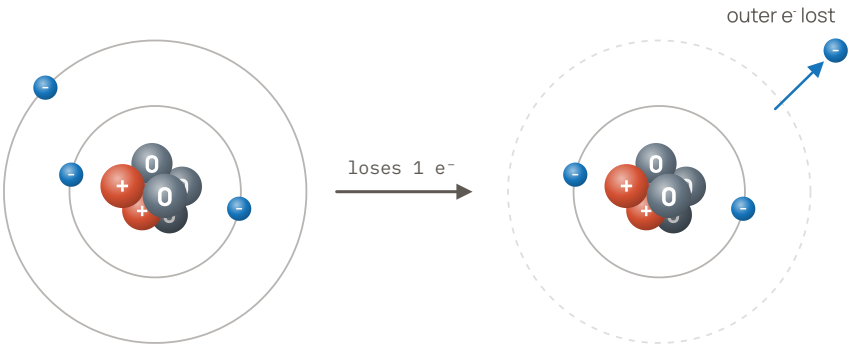
hydrogen-2 (deuterium)



1 p, 2 n

hydrogen-3 (tritium): unstable

Ions



lithium atom
3 p, 3 e → charge 0

lithium ion, Li^+
3 p, 2 e → charge +1

The outer electrons are furthest from the nucleus and are easiest to remove. If an atom **loses one or more outer electrons**, it has more protons than electrons, so it becomes a **positive ion**.

The nucleus does not change, so it is still the same element. (In chemistry you also meet negative ions, which form when atoms gain electrons.)

WORKED EXAMPLE

A magnesium atom, $^{25}_{12}Mg$, loses 2 electrons. Find the numbers of protons, neutrons and electrons, and the charge on the ion.

PROTONS	12 (unchanged)
NEUTRONS	25 − 12 = 13 (unchanged)
ELECTRONS	12 − 2 = 10
CHARGE	**+12 − 10 = +2, so Mg^{2+}**

YOUR TURN

An aluminium atom, $^{27}_{13}Al$, loses 3 electrons. Find the numbers of each particle and the charge.

PROTONS	13
NEUTRONS	14
ELECTRONS	13 − 3 = 10
CHARGE	+3, so Al^{3+}

PRACTISE

Try it

1 Are these two atoms isotopes of the same element? Circle **Yes** or **No** for each pair.

A $^{12}_{6}C$ $^{14}_{6}C$ — (Yes) No

B $^{14}_{6}C$ $^{14}_{7}N$ — Yes (No)

C $^{35}_{17}Cl$ $^{37}_{17}Cl$ — (Yes) No

D $^{40}_{18}Ar$ $^{40}_{20}Ca$ — Yes (No)

Explain your answer for pair B.

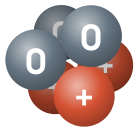
Different numbers of protons (6 and 7), so they are different elements, even though the mass numbers are the same.

2 SUPPORTED Chlorine has two common isotopes. Complete the table.

Isotope	Symbol	Protons	Neutrons	Electrons
chlorine-35	$^{35}_{17}Cl$	17	18	17
chlorine-37	$^{37}_{17}Cl$	17	20	17

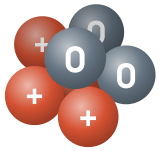
3 The diagrams show three nuclei. Which two are isotopes of the same element? Explain how you know.

P



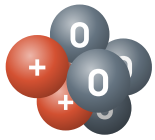
3 p, 3 n

Q



4 p, 3 n

R



3 p, 4 n

P and R: both have 3 protons (the same element, lithium) but different numbers of neutrons (3 and 4).

4 Complete the sentences.

When an atom loses an outer electron it becomes a ___positive___ ion. The number of ___protons___ does not change, so it is still the same ___element___. The ion has more ___protons___ than ___electrons___.

WORD BANK: positive | negative | protons | neutrons | electrons | element

5 ON YOUR OWN Complete the table. Write *atom* or *ion*, and the overall charge.

Protons	Neutrons	Electrons	Atom or ion?	Overall charge
11	12	11	atom	0
11	12	10	ion	+1
20	20	18	ion	+2
8	8	8	atom	0

6 Explain what is wrong with Ava's statement.

> **Ava:** A lithium atom becomes a Li^+ ion by losing a proton, so its charge goes up by one.

The atom loses an electron, not a proton; the nucleus does not change. Losing a negative electron leaves one more proton than electrons, so the charge is +1. If it lost a proton it would become a different element (helium).

7 CHALLENGE Uranium-235, $^{235}_{92}U$, and uranium-238, $^{238}_{92}U$, are both found in uranium ore. Explain why they are isotopes. Give one way their atoms are the same and one way they are different.

Same number of protons (92), so the same element, but different numbers of neutrons (143 and 146). Same: 92 protons / 92 electrons / same chemistry. Different: number of neutrons / mass number / how stable the nucleus is.

AQA-STYLE Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: 'Similarities and differences' needs both: give at least one of each, with numbers.

8 Carbon has two isotopes, $^{12}_{6}C$ and $^{14}_{6}C$. Describe the similarities and differences between the nuclei of these two isotopes. **[3 marks]**

Both have 6 protons (1). Carbon-12 has 6 neutrons and carbon-14 has 8 neutrons (1). So they have different mass numbers / carbon-14 is unstable (1).

9 Explain how a sodium atom can become a sodium ion, Na^+. **[2 marks]**

It loses one outer electron (1). It now has 11 protons and 10 electrons, so the overall charge is +1 (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Isotopes: same number of protons, different numbers of neutrons.
- Isotopes of an element have the same atomic number but different mass numbers.
- Some isotopes are unstable (radioactive), e.g. carbon-14.
- An atom that loses outer electrons becomes a positive ion.
- Forming an ion does not change the nucleus, so the element stays the same.

COVER THE PAGE AND ANSWER

1 What is the same in two isotopes of an element?

2 What is different in two isotopes of an element?

3 How does an atom become a positive ion?

4 An atom with 12 protons loses 2 electrons. What is its charge?

I CAN…

- explain what isotopes are
- use nuclear notation to compare isotopes
- describe how a positive ion forms
- work out the charge on an ion

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

REFERENCE

Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Ion: An atom that has lost (or gained) electrons, so it has an overall charge.

Isotopes: Atoms of the same element with different numbers of neutrons.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. 0
2. 31 − 15 = 16
3. It has the same number of electrons as protons.

COVER THE PAGE

1. The number of protons (atomic number).
2. The number of neutrons (mass number).
3. It loses one or more outer electrons.
4. +2